

From Homogeneous to Heterogeneous: Leveraging Deep Learning based Power Analysis across Devices

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Abstract—In this paper, we raise practical situations in profiling based power analysis when profiling and target devices are quite different at several levels. “Crossed devices” are newly termed, including homogeneous and heterogeneous devices, which have not been carefully investigated. We identify such device variations and take a further step towards leveraging the deep learning based power analysis. Traditional template attacks and straight-forward deep learning based power analysis will fail, when the gap across devices is significantly enlarged. In this paper, we propose a novel frequency and learning based power analysis mechanism, which is able to explore new attacking power of deep learning and address challenges caused by device variations. For the first time, power traces collected from our own PIC devices can be utilized to successfully attack the public dataset in DPAContest v4 which is based on a totally different AVR microcontroller.

I. INTRODUCTION

Profiling based side-channel attack (SCA) is a significant threat to embedded devices. An adversary can recover the secret key with the prior knowledge of the dependency between physical leakages and data being processed. The attack consists of two phases: *profiling* and *attacking*. In the former phase, an adversary collects physical leakages such as power consumptions or Electromagnetic Emissions (EM) from a profiling device which is a controlled copy of the target device, and builds templates or models. In the latter phase, the adversary can extract the secret key of the target device with power/EM traces collected from both devices. The most essential assumption is that the profiling device should be as similar to the target device as possible. Profiling based power attacks mainly include template-based power analysis [1], [2] (denoted as TA), machine learning-based power analysis [3]–[5] (ML-PA) and, recently, deep learning-based ones [6]–[9] (DL-PA).

DL-PA normally does not require pre-processing compared with ML-PA and TA, therefore a number of recent works are devoted to this area. Maghrebi et al [6] first proposed to use deep learning technology to build the classifier and recovered the secret key for an AES algorithm running on a microcontroller. Cagli et al [7] further explored the applicability of the convolutional neural network (CNN) to break the jitter based countermeasures. They combined CNN with the data augmentation technique in the case of over-fitting and trace misalignment, showing that it can effectively improve the efficiency of power analysis. Benadjila et al [8] implemented DL-PA on a masked AES with no prior knowledge of the mask value. Kim et

al [9] showed that even with some artificial noise, CNN can recover the key using only one trace.

Despite of the positive results from profiling based SCA, the concern about *device variations* has not been raised with enough attentions. In many laboratory experiments, power consumptions in both profiling and attacking phases are collected from the same device in one acquisition campaign. However, in practical scenarios, even though data are collected from the same device, they might still differ due to different ambient temperatures, environmental noises or measurement variations. It makes the key recovery still difficult on a single device, not to mention that on two physical copies.

The concept of *cross-device attack* refers to the scenario of using power traces collected from one device to attack the other one. However, in our opinion, the degree of (dis)similarity between profiling and target devices is still very vague. In this paper, we categorize such variation into four types. **(1) Same devices:** The traces used in both profiling and attacking phases are collected from the exact same device. The only difference is the key variation. **(2) Identical devices:** The profiling and target device are two physical copies of the same chip model. All the design and configurations are identical. This scenario is very common in laboratory experiments. Normally, the two identical copies are either purchased as off-the-shelf ones from the same batch, or built from the same circuit design. The slight variation may come from different custom printed circuit boards (PCB) [10] or different RAM size [11]. **(3) Homogeneous devices:** This case actually extends the dissimilarity of crossed devices quite a lot at the chip level. Both devices have chips from the same manufacturer but with different models and structures. **(4) Heterogeneous devices:** This is the ultimate problem in practice which is believed to be the most difficult to solve. The core chips of the two devices to be compared come from different manufacturers, which naturally differ in all aspects, such as models, instruction set architectures (ISA), power dissipations, etc. These four types are newly termed in this paper, which are used to clarify the level of the so-called “crossed devices” and to describe the extent of another attacking capability of DL-PA.

Some researchers became aware of such differences and conducted TA on two identical devices. Renauld et al [12] confirmed the existence of variation between profiling and target device so that TA may not succeed in practice. Lomné et al [13] conducted TA and linear regression analysis on several copies of microcontrollers with different CMOS technologies (350, 130 and 90nm). But they actually performed the attack only on different copies of the same model (i.e., *identical* devices in our terms), while the attack across devices of different models has not been fulfilled yet. Montminy et al [11] analyzed different devices with different RAM sizes in the same series (i.e., *identical* devices). They utilized TA with a specific preprocessing method (zero-mean unit-variance normalization) to eliminate the distribution difference between traces collected from

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different devices. Hanley et al [14] implemented a multi-device TA with pre-processing methods on 20 identical smartcards. Choudary et al [15] found that DC offset is one of the most important reasons for the low performance of TA. They manipulated the DC offset to improve TA across *identical* devices. Note that the objects that previous TA already addressed are mostly identical devices.

Recently, DL-PA and ML-PA are utilized to conquer the variation of devices. Das et al [16] first implemented DL-PA on the crossed devices and combined the multi-device training to further improve the performance. Multi-device training takes advantage of traces collected from multiple *identical* devices to train the model. Based on Das's work, Golder et al [17] introduced the principle component analysis (PCA) as a possible choice for cross-device attacks. Carbone et al [18] used DL-PA to recover the secrets of RSA running on an ARM Core SC 100. And the datasets for training, validation, and testing are collected from identical devices. Bhasin et al [19] stressed the importance of portability in SCA and implemented ML-PA and DL-PA to evaluate the performance of cross-device attacks. Overall, deep learning technique has demonstrated its potential in SCA. In many scenarios, it can enhance those statistical methods and reduce the number of traces that are required.

Note that all devices of the same/identical model meet their production standards, which means the variations among them are limited to a certain extent. In that case, we take one step further to exploit the possibility of DL-PA on homogeneous and heterogeneous devices and give our own proposal to improve the performance.

To better understand the device variation and the corresponding attacks, we categorize related works into newly termed scenarios as shown in Table I.

TABLE I: Related Works of Cross-device Attack.

Device Variation	Attacking Methods	References
Same devices	TA	[1]–[3]
	ML-PA	[3]–[5]
	DL-PA	[6]–[9]
Identical devices	TA	[11], [12], [14], [15]
	ML-PA	[13], [19]
	DL-PA	[10], [16]–[19]
Homogeneous devices	TA	Our work
	DL-PA	
Heterogeneous devices	TA	Our work
	DL-PA	

Contribution. Frequency analysis is another type of attack which gives a different angle and shows certain advantages. Our basic idea is to combine learning with analysis in frequency domain. The main contributions of our work include:

- We find two statistical metrics to quantitatively evaluate the (dis)similarity across different devices. To the best of our knowledge, it is the first time to identify the so-called homo/heterogeneous devices which have not been carefully explored before.
- We conduct comprehensive evaluations of different methods in the scenarios aforementioned. As for those state-of-the-art attacks, neither template attack nor deep learning-based power analysis can work well across homogeneous and heterogeneous devices.
- We propose a novel attack, called Frequency and Learning based Power Analysis (in short, FL-PA), which applies the deep learning to the frequency domain. Such attack leverages the new attacking power for DL-PA to conquer even those heterogeneous devices, which is assumed as the most challenging scenario of crossed devices with most variations.
- We present two case studies to validate our proposal. For the first time, DL-PA is applied to the public dataset in DPAContest v4 which is well-known in SCA community. We can use the power

traces collected from a local PIC device to attack the ATMega-163 in DPAContest v4. All experimental data can be publicly verified.

Organization. The paper is organized as follows: Sec.II gives the background. Sec.III shows the device variations. Sec.IV verifies the incapability of previous attacks. Sec.V and Sec.VI describe our attack and evaluations. Sec.VII concludes the paper.

II. PRELIMINARIES

A. Template Attack

Template attack (TA) [1] is a traditional method of profiling based SCA. The power consumptions of a device are related to the instructions and data being executed, and subject to a multivariate Gaussian distribution. In a typical trace-acquisition procedure, the instructions are fixed while data to be processed are changing due to the different inputs of plaintexts. TA builds a statistical model (known as templates) to describe the feature of power consumption for every possible value of data. More specifically, e.g., in an AES encryption, it calculates the intermediate value $v = Sbox(roundkey \oplus plaintext)$ of each trace and construct templates $(m, Cov)_v$, where m is the average of power traces with label v and the covariance matrix Cov represents the characteristics of power trace distribution with label v . Once the templates are built, in the attacking phase, the adversary collects power traces from the target device and calculates the probability of matching (between the templates and the power trace with specific key guess) on the basis of multivariate Gaussian distribution probability density function and Bayes' theorem. It is important to point out that in most cases, one trace does not have enough credibility to distinguish the correct key from all other candidates. A Maximum Likelihood Estimate (MLE) can be combined to facilitate the key prediction with multiple traces.

B. Deep Neural Network

The nature of TA can be viewed as a classification task for multi-classes. Deep neural network (DNN) has shown its advantage in this field, which uses plenty of training data and labels to approximate a complex function. A basic DNN consists of three parts: an input layer, several hidden layers and an output layer. A neuron is the basic unit of layers which implements a specific nonlinear function with several inputs and one output. The input of each neuron is the linear combination of the outputs of those in the previous layer. A DNN model for classification also has two phases. In the training phase, it builds a model with a specific structure and adjusts the model parameters with training data through back propagation to make classification predictions more accurate. In the classification phase, it outputs a probability for each class. The one with the highest probability is the prediction result of the network.

C. DPAContest v4

DPAContest v4 is an open dataset that provides power measurements of a masked implementation of AES on an ATMega-163 smartcard from ATMEL. There are two versions: v4.1 and v4.2. The datasets are acquired on a standard evaluation board called SASEBO-W. All the details are available online [20], including implementations, acquisitions and reference traces with keys, plaintexts, offsets, etc. More details about DPAContest v4 will be discussed in Sec.VI-B.

III. VARIATION IN CROSSED DEVICES

The variation between the profiling and target devices actually exists. In fact, such variation is not easy to overcome. For example, PIC from Microchip and AVR from ATMEL*, both differ in most of

* We are aware that ATMEL was acquired by Microchip Technology Inc., but the dissimilarity between them is very huge.

fundamental architectures including instruction sets, clock cycles per instruction, stack implementations, etc.

In this paper, several devices are investigated in our experiments and listed in Table II. Each ID indicates a specific model. The ID with suffixes is a copy of that model. In Table II, M163-1 and M163-2 are two identical devices. M163 and M32L are considered as homogeneous while M16A and P887 are heterogeneous.

Note that the (dis)similarity of crossed devices is already clarified, however, the degree of such difference has not been evaluated. In this paper, we first propose two methods to verify such variations at both theoretical and practical levels.

TABLE II: Illustrative Crossed Devices with Different Models and Instruction Set Architectures (ISA).

ID	Model (ATMega)	ISA	ID	Model (PIC)	ISA
M8515	8515-16AU	AVR4	P887	16F887	Midrange
M163	163L-4AC	AVR5	P914	16F914	Midrange
M16A	16A-AU	AVR5	PK80	18F45K80	High-Performance
M32L	32L-8AU	AVR5	PK22	18F45K22	High-Performance
M324	324PA-AU	AVR5			
M644	644-20AU	AVR5			

A. Variation at the Theoretical Level

Since the power traces follow a multivariate Gaussian distribution, two mathematical metrics, *Wasserstein Distance* (WD) [21] and *Maximum Mean Discrepancy* (MMD) [22], are adopted for the first time to quantify the numeric variations of devices.

Wasserstein Distance, also known as KantorovichRubinstein distance, is a good metric used to quantify the difference between two multivariate Gaussian distributions. It measures the minimum value of the average distance when “moving” the data from one distribution to the other. Recently WD is used in Generative Adversarial Networks (GAN) to improve the performance [23].

Maximum Mean Discrepancy is a distance in the space of probability measures, which has numerous applications in machine learning and nonparametric testing. It is based on the notion of embedding probabilities in a reproducing kernel Hilbert space. MMD is used in transfer learning to measure the difference between two datasets [24].

For each device, 10k traces for the same AES-128 implementation are collected. Results of two metrics are shown in Fig.1a and Fig.1b, respectively. Note that both distances are normalized to the range of [0,1] which is colored from pure white to dark blue. The observations of device variations can be briefly summarized as followings:

- The same device scenario is along the diagonal of both Fig.1a and Fig.1b (from top left to bottom right), which has minimal distance (marked in pure white).
- The cases of identical devices of M163-1/2 and P887-1/2 (sized 2×2) are closely adjacent to the diagonal, showing that the distance across such cases is still small. It is more distinct in Fig.1b.
- The homogeneous devices correspond to two squares (sized 7×7 top left and 5×5 bottom right). According to the colors marked in each square, the mutual distances for devices from the same manufacturer are relatively small, indicating that profiling based attacks are still possible.
- The pairwise distances between any AVR and PIC devices are apparently very large. The two rectangles (sized 5×7) with dark blue shown in Fig.1b imply that the cross device attack should be very difficult and might be impossible.

The results in Fig.1a and Fig.1b depict the degree of (dis)similarity among different types of crossed devices that we define, which is theoretically verified through variations represented in colors.

B. Variation at the Practical Level

Correlation power analysis (CPA) [25] is a standard power analysis technique. It calculates the correlation between the hypothesis and real traces using power models such as Hamming Weight (HW) or Hamming Distance (HD) model. Some distinct peaks will appear in the correlation curve, which correspond to the most-likely key hypothesis compared with those for incorrect key guesses. For a target device (without SCA countermeasures), a successful CPA can reveal the exact timing of a specific operation, which can be used to detect and evaluate the device variation from another point of view.

For each device, the first 500 traces among those 10k collected in Sec.III-A are reused. The target operation is the first SubBytes in the first round of AES encryption. The first byte of secret key is successfully recovered for all devices through a standard CPA. The correlation coefficients curve of each device for the correct key guess are shown in Fig.1c. x -axis stands for discrete-time points and y -axis shows devices of different models. The depth of red color corresponds to the absolute value of each correlation coefficient along the curve, ranging in [0,0.8].

Note that all traces are triggered and well-aligned at the time point 0. For AVR devices, the peak of the correlation coefficient appears quite earlier and it varies in a narrow region, i.e., point range [1k, 2k]. For the PIC device, the target SubBytes operation is executed quite later, which varies in a broader range of [8k,10k]. This is due to the different instruction sets for midrange and high-performance devices where the assemblies for the same C code are quite different.

From another perspective, in Fig.1c, the characteristics of power patterns are similar with those identical devices, such as M163-1/2, P887-1/2, and the pair of M644 and M324. The latter pair is from the same series with different RAM sizes. In contrast, heterogeneous devices such as M163 and PK22 show a large variation across them.

IV. BASELINE TEMPLATE-BASED AND DL-BASED POWER ANALYSIS ON HOMO/HETERO-GENEOUS DEVICES

In Sec.III, the variation indicates that it may be more difficult for profiling based power analysis to attack homo/hetero-geneous devices than same/identical devices. It also implies that previous TA or DL-PA may not work in homo/hetero-geneous cases, which will be verified in this section.

A. Experiment Setup

Our acquisition platform is based on SAKURA-W [26] which is a successor of SASEBO-W that is used in DPAContest v4. Customized PCB boards for device chips are designed to fit the interface slot. The clock of devices is locked at 3MHz provided by SAKURA-W. The source code we used is written in C which implements an unprotected AES-128. A few necessary adjustments are made on the IO pins to make sure that the devices work. The encryption module for all devices remains unchanged, empowering us to focus on the device variation merely. The C code is then compiled by arm-gcc (for AVR series) and XC8 (for PIC series), and the compiled binaries are uploaded to the devices. In order to focus on device variation rather than key variation, the same key is fixed when sent as inputs together with plaintexts. The oscilloscope is KeySight DSOX3034T with a sampling rate of 1.25GSa/s. All the traces are pre-aligned with the same trigger at the beginning of the encryption. 16k traces are collected for each device. Among them, 500 consecutive points are chosen as our data which will cover the first SubBytes operation. The value of points ranges in [0,255].

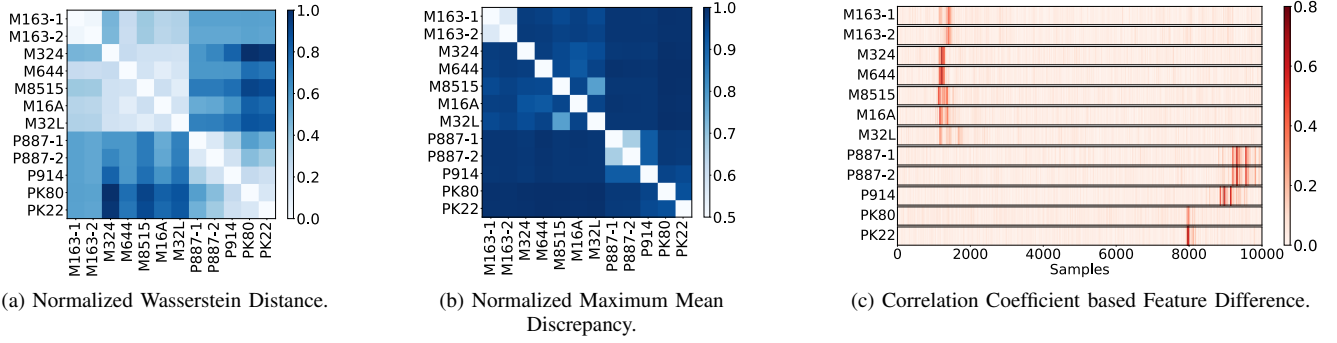


Fig. 1: Device Variations for AVR and PIC at the Theoretical and Practical Levels.

B. Template-based Power Analysis (TA)

A template attack is implemented and served as the baseline for comparisons. In the profiling phase, 10 points of interest (POI) are selected which correspond to the largest correlation of power consumption for the first SubBytes operation. The validity of POI can also be verified through a CPA around the largest correlation coefficient. In the attacking phase, 10 points are also chosen from power traces collected in the target device, which have the same index as that in the profiling phase. A maximum of 1k traces are used for attacking all crossed devices. In this paper, in order to evaluate different crossed devices, we define the metric as the number of the traces required for a full recovery of key byte (i.e., the guessing entropy keeps 0). The results are shown in Fig.2a. The depth of the color represents the number of traces used for key recovery.

It is easy to observe that TA can recover the key for same and identical devices (on the diagonal). For same devices, all the keys can be recovered within 3 traces. For identical devices, the number of traces required in the attacking phase is less than 30. When turning to homogeneous devices, those TAs can still work well on PIC with a small amount of traces, as shown in the square sized 5×5 bottom right. However, it will be difficult for TA to recover keys in AVR, as shown in the square sized 7×7 top left. Some AVR devices fail to recover the key for other AVR devices, due to the different POI regions of profiling and target devices as shown in Fig.1c. More importantly, it will be nearly impossible for launching TA on heterogeneous devices across AVR and PIC (within 1k traces), which is shown in the dark red regions located at top right and bottom left in Fig.2a.

C. Deep Learning-based Power Analysis (DL-PA)

Instead of selecting those POI, the entire dataset from the profiling device are fed as the training data, therefore the input size of the

neural network is fixed to 500. A 256-class DNN with two hidden layers is implemented. The results are shown in Fig.2b.

For identical devices, DL-PA takes less than 15 traces to recover the key. In PIC series, it has a generally better performance across devices than that in AVR series. Similar to TA, DL-PA also meets a great difficulty when attacking heterogeneous devices. In Fig.2b, the two big rectangular regions with all dark red indicate that a simple implementation of learning-based attacks may not work directly, thus a new technique to conquer this problem is in demand.

V. OUR ATTACK

According to our empirical experience, the failure of template attacks on homo/hetero-geneous devices is attributed to the following factors: selections of POI region and cycles of instruction. Both are closely related to the clock in time domain. In nature, learning-based power analysis has a stronger generalization capability than those statistical analyses such as TA. To improve DL-PA and mitigate those issues in time domain, one important motivation yet also a reasonable practice is to migrate deep learning into the frequency domain. In this paper, we propose our new attack named *Frequency and Learning based Power Analysis*, in short FL-PA, which is considered as one of the most significant contributions of the proposed work.

A Fast Fourier transform (FFT) is applied to all power traces from those crossed devices. Fig.3a shows an exemplary trace after FFT for one specific AVR device on the bottom and its CPA results in frequency domain on the top. In comparison, Fig.3b lists the corresponding results for a PIC device. Both of the correlation coefficient curve and the frequency spectrum in Fig.3a are significantly different from those in Fig.3b. One important reason lies in the differences in clock cycles and instruction cycles between AVR and PIC. Hence, a direct implementation of a DL-PA in the frequency domain across heterogeneous devices would also fail, as similar to results in Sec.IV.

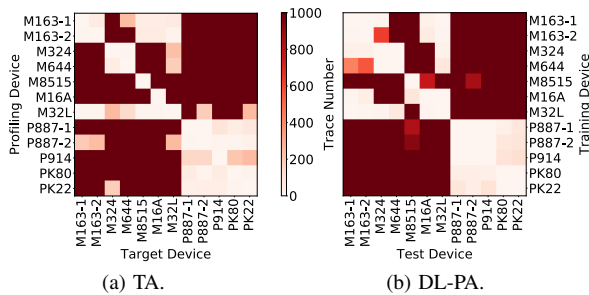


Fig. 2: Results of Previous Attacks across Devices.

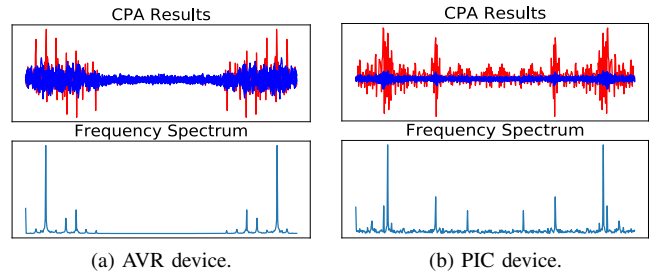


Fig. 3: Traces after FFT and Correlation Coefficient.

The frequency spectrum of a power trace can be viewed as a sparse matrix. And there are several spikes which are easy to observe and they are more likely associated with the data being processed. Similar to the selection of POI region in time domain, it is also important to apply such selection in frequency domain. In our FL-PA, we select the frequency components with the highest magnitudes exclude the DC offset (whose point index is 0) as the characterization of frequency. It is equivalent to applying a customized bandpass filter to each trace and extracting the interested features with reduced noise. Due to the symmetry of FFT, only the frequency components in the first half of the frequency spectrum are selected. As a result, 7 out of 500 points are used for further analysis.

The neural network architecture of our FL-PA is shown in Fig.4. The activation function for input and hidden layers is Rectified Linear Unit (ReLU), and that for output layers is softmax with a cross-entropy loss. The dropout layer and L_2 regularization are implemented in hidden layers to avoid over-fitting. In back propagation, the root mean square propagation (RMSprop) with 0.00001 learning rate is adopted, which has been proved as the best optimizer in SCA [8]. 10k traces from one specific profiling device are used to train the model. 5k traces for each device in Table II are used during attacks. The model is trained for 100 epochs with a batch size of 100, where the early stopping technique is used to avoid over-fitting. All the data has been normalized before entering the neural network.

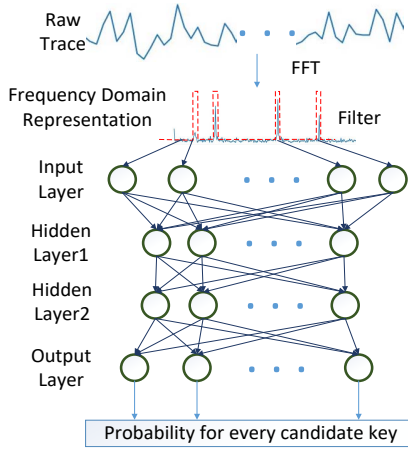


Fig. 4: Neural Network Model Architecture.

VI. EVALUATION

To evaluate the proposed FL-PA attacks, two case studies are explored. One is on the local crossed devices. The other is on the public dataset from DPAContest v4, which is quite well-known and open to the SCA community. Our experimental results can be reproduced and publicly verified.

A. Case Study 1: Attacks on Local Datasets

In this study, AES-128 is implemented on all local devices. For each attack, one device is served for both profiling and training, and the other is for attacking. Both traces are collected by our own scope. For each device, only one unified model is trained yet will be applied to attack all crossed devices.

The attack results of FL-PA across different devices are shown in Fig.5a. Within 600 traces, all attacks can succeed on all pairs of crossed devices. (1) For those heterogeneous devices (AVR v.s. PIC), compared with TA and DL-PA results in Fig. 2, those rectangular

regions with dark red are now turning to light colors. Hundreds of traces are enough to recover the key of a PIC device using AVR. (2) For those homogeneous devices, those square regions (sized 5×5 and 7×7) are showing a color much lighter. For any pair by the same manufacturer (either AVR or PIC), the number of traces required in our FL-PA is merely less than one hundred, which is better than that in Fig.2b. (3) For those same/identical devices, due to the techniques we used to prevent over-fitting, the performance of our attack has deteriorated. For example, M324 requires 10 traces for FL-PA compared to 2 for DL-PA. The explanation is the so-called “No Free Lunch” theory. The model we trained is generic and could adapt to all crossed devices, not just a specific one only for M324.

The loss on test data can well reflect the generalization ability of the model. The loss of our FL-PA compared with DL-PA in Sec.IV on crossed devices are shown in Fig.5b and Fig.5c. The training device is M163-1. x -axis and y -axis are the number of epochs and the loss value during model training, respectively. For DL-PA in Fig.5b, M163-1 (same device) and M163-2 (identical device) have relatively small losses. For other homo/hetero-geneous devices, their losses are getting even larger after 20 epochs, which can be considered as another form of over-fitting. This shows that DL-PA in the time domain does not learn robust features from physical devices. For our FL-PA shown in Fig5c, the loss for M163-1 remains minimal. Meanwhile, the losses for all other devices also converge and are very close to M163-1. Such results confirm that our model in the frequency domain has better capability and strong robustness for generalization across different devices.

B. Case Study 2: Attacks on Public Dataset (DPAContest v4)

In this study, only one device is under our control and served for training. The target device is chosen as ATMega-163 in DPAContest v4, which is out of our reach. More specifically, we focus on v4.1 which is a software implementation of AES-256 with Rotating SBoxes Masking countermeasure (RSM). The trace dataset of v4.1 is publicly well-known and already acknowledged by the SCA community. Details about v4.1 are available online [20]. In practice, it is quite difficult to find the exact same device with the exact same acquisition equipment and measurement parameters, which makes our investigation on DPAContest v4.1 quite challenging. Table III lists the difference of acquisition and device setting between our PIC devices and the ATMega-163 used in the public dataset. The apparently disjoint settings in Table III can be considered as the ultimate case for crossed devices. This should be the most difficult to attack, resulting in the failure of TA and DL-PA as shown in Sec.IV. However, our FL-PA can manage to conquer such heterogeneity.

TABLE III: Acquisition and Device Variation between our PIC Microcontrollers and ATMega-163 in DPAContest v4.

Settings	Local	DPAContest v4.1
Chip Model	PIC16F887	ATMega-163
	PIC16F914	
	PIC18F45K80	
	PIC18F45K22	
Cipher	AES-128	Masked-AES-256
Target Operation	Subbytes	Masked-Subbytes
Sampling Rate	1.25GS/s	500MS/s
Oscilloscope	KeySight DSOX3034T	Lecroy Waverunner 6100A
Platform	SAKURA-W	SASEBO-W

We implement FL-PA and train the model from local PIC devices to attack the public dataset in DPAContest v4.1. As a common practice in the community, the value of masks is assumed as known to the adversary. Hence the AES implementation in v4.1 is equivalent to a standard AES with the prior knowledge of mask value. Recall that the main purpose of FL-PA is to leverage the capability of learning based

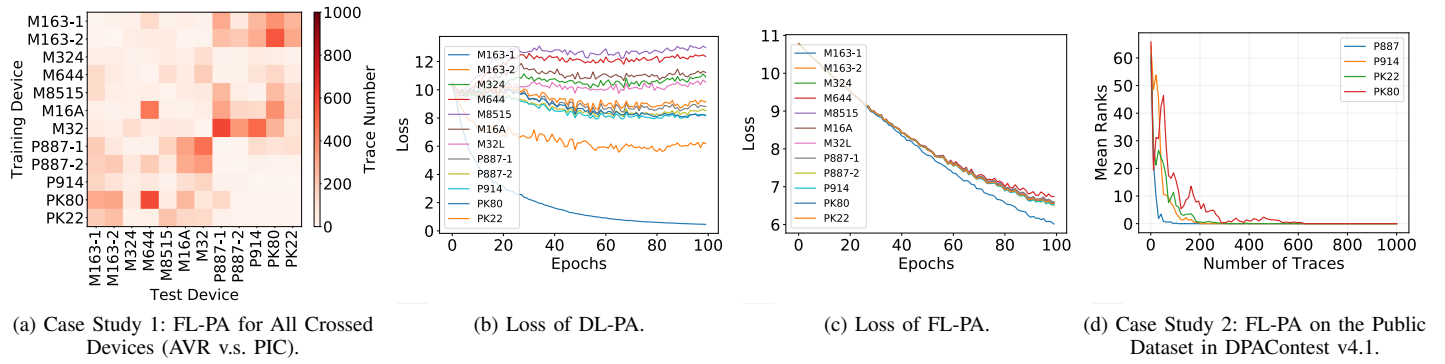


Fig. 5: Case Studies and Evaluation Results of our Frequency and Learning based Power Analysis on Crossed Devices.

power analysis across heterogeneous devices, instead of defeating SCA-countermeasures. Therefore, the equivalent dataset from v4.1 is a good fit for our illustration.

In our attack, the training dataset is the same as that in Sec. VI-A. As for the test dataset, 500 consecutive points [101200, 101700] are truncated from the raw traces of the public dataset in DPAContest v4.1 and served as the target dataset. After applying FFT, all local maximum points are extracted. Among them, the first 7 points are selected as the inputs of neural network for each trace. The target intermediate value is the first output byte of a masked Subbytes operation, where $v = Sbox(roundkey \oplus plaintext) \oplus mask$.

Fig. 5d shows the attack results of recovering the first key byte for the AES-256 in DPAContest v4.1. x -axis is the number of the traces that are required for key recovery. y -axis is the guessing entropy for the correct key guess, i.e. the remained key search space. Using all PIC devices listed in Fig. 5d, FL-PA can succeed in extracting the key of ATMega-163 within 800 traces, indicating its robustness for crossed devices. Note that the success of our attacks conquered multiple factors of variations in practice, including device, cipher implementation, sampling rate, etc.

VII. CONCLUSIONS

In this paper, we put insights into the boundary of crossed devices and categorize them into four types: same, identical, homogeneous, and heterogeneous devices. Most importantly, we leverage the new attacking power of the frequency and deep learning based power analysis to conquer the so-called device variations, from homogeneous to heterogeneous devices. Our research bridges the gap of device variation and makes it possible to use traces collected from one device to attack a totally different one, such as AVR v.s. PIC.

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